Derivation of Pharmacokinetic Rate Constants from Volumes and Clearances

Introduction

In compartmental pharmacokinetic models, drug distribution and elimination are described using rate constants that govern the movement of drug between compartments and out of the system. These rate constants are derived from physiological parameters such as compartment volumes and intercompartmental clearances.

Compartment Definitions

We consider a three-compartment model:

- Central compartment: volume V_1
- Peripheral compartment 1: volume V_2
- Peripheral compartment 2: volume V_3

Intercompartmental clearances:

- Q_2: clearance between central and peripheral compartment 1
- Q_3: clearance between central and peripheral compartment 2

Systemic clearance from the central compartment is denoted as Cl.

Rate Constant Derivations

The rate constants are defined as follows:

Elimination Rate Constant

$$k_{10} = \frac{Cl}{V_1}$$

This represents the rate of drug elimination from the central compartment.

Intercompartmental Rate Constants

$$
\begin{aligned}
k_{12} &= \frac{Q_2}{V_1} \quad \text{(central to peripheral 1)} \\
k_{21} &= \frac{Q_2}{V_2} \quad \text{(peripheral 1 to central)} \\
k_{13} &= \frac{Q_3}{V_1} \quad \text{(central to peripheral 2)} \\
k_{31} &= \frac{Q_3}{V_3} \quad \text{(peripheral 2 to central)}
\end{aligned}
$$

These equations assume that the intercompartmental clearance Q is symmetric and that the rate constants are first-order with respect to drug concentration.

Units

- Clearance (Cl, Q_2, Q_3): L/min or mL/kg/min
- Volume (V_1, V_2, V_3): L or L/kg
- Rate constants (k_{ij}): min^{-1}

Conclusion

By combining compartment volumes and intercompartmental clearances, we can derive the rate constants that define drug movement and elimination in a multi-compartment pharmacokinetic model. These constants are essential for simulating drug concentration-time profiles and for designing dosing regimens.